

Tianrui Qi

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EDUCATION

Ph.D. in Biomedical Engineering 09/2025 - 05/2030

Boston University, Boston, MA

B.S. in Computer Science 01/2023 - 05/2025

Georgia Institute of Technology, Atlanta, GA

- GPA: 3.85/4.00
- Achievements: Highest Honor, President's Undergraduate Research Awards

B.S. in Computer Science; Double Major in Mathematics 09/2020 - 12/2022

Rensselaer Polytechnic Institute, Troy, NY

- GPA: 3.73/4.00
- Achievements: Dean's Honor List (every semester)
- Minor: Economics

PUBLICATIONS

†: contributed equally

**Bio-CM²: A Computational Miniature Mesoscope
Enabling Cortex-Wide Cellular Imaging in Mice** 2026

Manuscript in Preparation

Guorong Hu, Tianrui Qi, et al., Lei Tian

Volumetric Localization Microscopy with Deep Learning 2025

Nature Communications

Keyi Han[†], Xuanwen Hua[†], Tianrui Qi[†], Zijun Gao, Xiaopeng Wang, Shu Jia

DOI: doi.org/10.1038/s41467-025-65941-3

Superiority of Quadratic Neural Networks for Classification of Gaussian Mixture Data 2022

Visual Computing for Industry, Biomedicine, and Art

Tianrui Qi, Ge Wang

DOI: doi.org/10.1186/s42492-022-00118-z

RESEARCH EXPERIENCE

Graduate Research Assistant 11/2025 - 07/2026

Boston University, Boston, MA

Computational Imaging Systems Lab, PI: Lei Tian, Ph.D.

- Analyzed reconstructed in vivo calcium-imaging videos covering a 10 mm × 7.5 mm field of view (FOV) from head-fixed mice to support biological validation of the computational miniature mesoscope (CM²).
- Applied Constrained Nonnegative Matrix Factorization for microendoscopic data (CNMF-E) to extract 3,474 neuronal footprints and corresponding 180-second calcium-activity traces from background-rich videos.
- Validated neuron extraction quality using spatial and temporal metrics, showing that the single action potential (AP) response kinetics of the GCaMP6s calcium indicator matched published values.

Graduate Research Assistant 03/2026 - 04/2026

Boston University, Boston, MA

Dunlop Lab, PI: Mary Dunlop, Ph.D.

- Developed an end-to-end time-series deep learning model to predict light-stimulation schedules that drive CcaS/CcaR optogenetic gene expression to follow target trajectories.

- Characterized model performance across simulation stochasticity, input-history lengths, target-trajectory dynamics, and model architectures, identifying factors driving trajectory-tracking error.

GitHub: github.com/tianrui-qi/GEPC

Graduate Research Assistant

09/2025 - 12/2025

Boston University, Boston, MA

Biomedical Optical Technologies Lab, PI: Darren Roblyer, Ph.D.

- Developed a transformer-based cuffless blood pressure (BP) estimation pipeline driven by speckle contrast optical spectroscopy (SCOS), with flexible inputs handling variable-length and missing data.
- Established a three-stage training pipeline consisting of unsupervised representation learning on BFi and PPG waveforms, supervised regression on BP waveforms, and subject-specific finetuning.
- Visualized representations after unsupervised pretraining on optical waveforms, showing that subject-level differences were captured across measurements and that BP-related patterns were encoded.

GitHub: github.com/tianrui-qi/SCOS-BP

Undergraduate Research Assistant

04/2023 - 05/2025

Georgia Institute of Technology and Emory University, Atlanta, GA

Jia Laboratory for Systems Biophotonics, PI: Shu Jia, Ph.D.

- Engineered a scalable 3D U-Net pipeline based entirely on simulated data for volumetric localization in single-molecule super-resolution microscopy, resolving sub-cellular structure down to 60 nm.
- Developed a patch-based prediction pipeline that flexibly adapts to various input volume sizes and achieves a 100x speedup over conventional deterministic localization methods.
- Integrated the redundant cross-correlation algorithm for drift correction with the deep learning-based prediction pipeline.

GitHub: github.com/tianrui-qi/DSMLM

Undergraduate Research Assistant

11/2021 - 12/2022

Rensselaer Polytechnic Institute, Troy, NY

AI-based X-ray Imaging System Lab, PI: Ge Wang, Ph.D.

- Derived forward and backward propagation formulations for conventional and quadratic neural networks, with from-scratch NumPy implementation of full neural-network training workflows in Python.
- Demonstrated that single-layer quadratic neural networks rival conventional neural networks with hundreds of neurons in classifying simulated and real-world Gaussian mixture data.

GitHub: github.com/tianrui-qi/QuadraticNeurons

INDUSTRY EXPERIENCE

Co-op

01/2024 - 08/2024

Regeneron Genetics Center, Tarrytown, NY

Therapeutic Area Genetics, Manager: Jing He, Ph.D.

- Utilized BERT-based large language models (LLMs) and unsupervised feature selection to obtain a vector representation in a bio-meaningful space for each whole exome sequencing (WXS) sample.
- Demonstrated that the representations capture sample-wise differences by predicting immune system indicators of The Cancer Genome Atlas Program (TCGA) skin cancer samples.
- Scaled up the pipeline to handle hundreds WXS samples with billion DNA sequences by optimizing parallel computing for high-performance computing (HPC) and enhancing file system efficiency through hashing.

GitHub: github.com/tianrui-qi/SIP-DB2

TEACHING EXPERIENCE

Undergraduate Teaching Assistant

09/2022 - 12/2022

Rensselaer Polytechnic Institute, Troy, NY

CSCI 2200 Foundations of Computer Science, Instructor: David Goldschmidt, Ph.D.

SKILLS

Programming: Python (PyTorch, Lightning, NumPy, pandas), MATLAB, Java, C, C++, R, Bash, MIPS.

Development Tools: Git, Conda, VS Code, RStudio, JetBrains Suite, Android Studio.

Computing Platforms: Linux (Ubuntu), AWS (EC2, S3), HPC (Slurm).

Software: Adobe Illustrator, ImageJ, LaTeX, OpenAI Codex.

Laboratory: optics and laser alignment, fluorescence imaging, fluorescence labeling, cell culture maintenance.

Communication: English (Professional), Mandarin (Native).

COURSE PROJECTS

WanderSync: A Collaborative Travel Management System

09/2024 - 12/2024

Georgia Institute of Technology, Atlanta, GA

CS 2340 Objects and Design, Instructor: Nimisha Roy, Ph.D.

- Developed a travel management application in Java, following the MVVM architecture and using Firebase Realtime Database to manage user accounts, destinations, accommodations, and dining reservations.
- Enabled real-time collaboration features, allowing users to seamlessly share and synchronize travel plans, including itineraries and notes, across group members.

GitHub: github.com/tianrui-qi/WanderSync

Datapath and Control for a Turing Complete ISA with Interrupt Handling

09/2023 - 12/2023

Georgia Institute of Technology, Atlanta, GA

CS 2200 Systems and Networking, Instructor: Daniel Forsyth

- Designed a single-bus datapath and an efficient four-ROM microcontroller for a Turing complete instruction set architecture (ISA), supporting arithmetic, logical, memory access, and control flow instructions.
- Handled basic and input device interrupts by additional hardware including new instructions, interrupt registers, signals, and I/O bus, along with software supports such as interrupt vector tables.

GitHub: github.com/tianrui-qi/LC2222a-ISA

Alternating Direction Method of Multipliers for Support Vector Machine

01/2022 - 05/2022

Rensselaer Polytechnic Institute, Troy, NY

MATP 4820 Computational Optimization, Instructor: Yangyang Xu, Ph.D.

- Formulated the primal and augmented dual optimization problems for support vector machine (SVM) objective and developed alternating direction method of multipliers (ADMM) solver.
- Implemented the ADMM solver in MATLAB and reported the primal and dual feasibility violations at each outer iteration for the testing datasets.

GitHub: github.com/tianrui-qi/ADMM-for-SVM

Windows of Susceptibility Analysis for Brain Diseases

01/2022 - 05/2022

Rensselaer Polytechnic Institute, Troy, NY

MATP 4400 Data Mathematics, Instructor: Kristin Bennett, Ph.D.

- Performed the windows of susceptibility analysis based on mouse data from a similar brain-in-a-dish model for mice using R with k-means clustering and principal component analysis (PCA).
- Analyzed the same sets of microcephaly-associated genes and Zika-associated genes and detected similar windows of susceptibility for Microcephaly and Zika-induced microcephaly in mice as in humans.

GitHub: github.com/tianrui-qi/WOS-Analysis